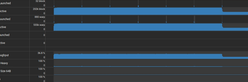
	Input	Output	Size	Time	Cycles	GPU	Process	Attributes
MLLA	mlp_2.0.exe	mlp_2.0.exe_0.pix	2.01 Gb	2:06:26	7,606,176,000	0 - NVIDIA GeForce RTX 5090 Laptop GPU	1.00 GHz	[0x0b] mlp_2.0.exe_Capturing_a_wait_period
Unreal	ue_5.exe	ue_5.exe_0.pix	2.01 Gb	0:01:00	1,000,000,000	0 - NVIDIA GeForce RTX 5090 Laptop GPU	1.00 GHz	[0x0b] ue_5.exe_Capturing_a_possible_scenario_waiting_event
Unreal	ue_5.exe	ue_5.exe_1.pix	2.01 Gb	0:01:00	1,000,000,000	0 - NVIDIA GeForce RTX 5090 Laptop GPU	1.00 GHz	[0x0b] ue_5.exe_Capturing_a_possible_scenario_waiting_event

Capture Settings | Analysis | Target | Capture | Comments | Page | Session
Capture Settings: Capture Type: Frame Capture | Capture Method: Automatic | Capture Time: 0.000 seconds | Capture Frame: 1

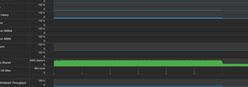
GPU Speed-of-Light (SOL)

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput of various pipelines. Memory is shown in a dedicated section. Throughput is reported at the SM, L1/TEX cache, L2 cache, and DRAM/HBM units. Compute and memory throughput are the GPU counterpart to a roofline chart.

Compute Throughput	93.94 %	2.82 GHz
Memory Throughput	93.94 %	14.000 GHz
L1/TEX Cache Throughput	93.97 %	2,010.41 GB/s
L2 Cache Throughput	93.94 %	1,609.60 GB/s
Device Memory Throughput	93.94 %	1,609.60 GB/s
DRAM Throughput	93.94 %	1,609.60 GB/s
Compute (SM) Throughput	93.94 %	2,010.41 GB/s
Memory Throughput	93.94 %	1,609.60 GB/s



GPU Speed Of Light (SOL) [%]



Pc Sampling

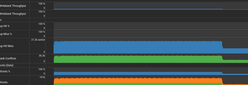
Total cycles: 7,606,176,000 (Samples aggregated proportionally over the workload duration. Data is collected across multiple kernel launches.)

Maximum Sampling Interval [cycles]: 2 (16.00% of 8192)
Maximum Buffer Fill: 256 (3.13% of 8192)

Pc Sampling	16.00 %	2
Stalled	16.00 %	2
Not Selected	16.00 %	2
Selected	16.00 %	2
Issue Active	16.00 %	2
Instruction Issued	16.00 %	2
Warp Cycles Per Issued Instruction	16.00 %	2
Avg. Not Selected Warps	16.00 %	2
Avg. Active Warps	16.00 %	2
Avg. Stalled Warps	16.00 %	2
Stall Barrier	16.00 %	2
Stall Barrier Sleep	16.00 %	2
Stall Barrier Wait	16.00 %	2
Stall Bed	16.00 %	2
Stall Bed Sleep	16.00 %	2
Stall Bed Wait	16.00 %	2
Stall Dispatch	16.00 %	2
Stall Dispatch Wait	16.00 %	2
Stall Dispatch Sleep	16.00 %	2
Stall Math Pipe	16.00 %	2
Stall Math Pipe Throttle	16.00 %	2
Stall Tex	16.00 %	2
Stall Tex Throttle	16.00 %	2
Stall Tex Mem Throttle	16.00 %	2
Stall Tensor	16.00 %	2
Stall Tensor 64Bit	16.00 %	2
Stall Tensor 64Bit Throttle	16.00 %	2
Stall Misc	16.00 %	2
Stall Misc Sleep	16.00 %	2
Stall Misc Wait	16.00 %	2
Stall Lsu	16.00 %	2
Stall Lsu Sleep	16.00 %	2
Stall Lsu Wait	16.00 %	2
Stall Lsu Mem Throttle	16.00 %	2
Stall Lsu Mem Throttle Global	16.00 %	2
Stall Lsu Mem Throttle Shared	16.00 %	2
Stall Lsu Mem Throttle Local	16.00 %	2
Stall Lsu Mem Throttle Other	16.00 %	2
Stall Lsu Mem Throttle Tex	16.00 %	2
Stall Lsu Mem Throttle Tex Global	16.00 %	2
Stall Lsu Mem Throttle Tex Shared	16.00 %	2
Stall Lsu Mem Throttle Tex Local	16.00 %	2
Stall Lsu Mem Throttle Tex Other	16.00 %	2
Stall Tex Mem Throttle	16.00 %	2
Stall Tex Mem Throttle Global	16.00 %	2
Stall Tex Mem Throttle Shared	16.00 %	2
Stall Tex Mem Throttle Local	16.00 %	2
Stall Tex Mem Throttle Other	16.00 %	2
Stall Tex Mem Throttle Tex	16.00 %	2
Stall Tex Mem Throttle Tex Global	16.00 %	2
Stall Tex Mem Throttle Tex Shared	16.00 %	2
Stall Tex Mem Throttle Tex Local	16.00 %	2
Stall Tex Mem Throttle Tex Other	16.00 %	2
Stall Sleep	16.00 %	2
Stall Sleep Sleep	16.00 %	2
Stall Sleep Wait	16.00 %	2
Stall Wait	16.00 %	2
Stall Wait Sleep	16.00 %	2
Stall Wait Wait	16.00 %	2
Stall Other	16.00 %	2
Stall Other Sleep	16.00 %	2
Stall Other Wait	16.00 %	2
Stall Sass	16.00 %	2
Stall Sass Sleep	16.00 %	2
Stall Sass Wait	16.00 %	2
Stall Sass Mem Throttle	16.00 %	2
Stall Sass Tex	16.00 %	2
Stall Sass Tex Throttle	16.00 %	2
Stall Sass Tex Mem Throttle	16.00 %	2
Stall Sass Misc	16.00 %	2
Stall Sass Misc Sleep	16.00 %	2
Stall Sass Misc Wait	16.00 %	2
Stall Sass Lsu	16.00 %	2
Stall Sass Lsu Sleep	16.00 %	2
Stall Sass Lsu Wait	16.00 %	2
Stall Sass Lsu Mem Throttle	16.00 %	2
Stall Sass Lsu Mem Throttle Global	16.00 %	2
Stall Sass Lsu Mem Throttle Shared	16.00 %	2
Stall Sass Lsu Mem Throttle Local	16.00 %	2
Stall Sass Lsu Mem Throttle Other	16.00 %	2
Stall Sass Tex Mem Throttle	16.00 %	2
Stall Sass Tex Mem Throttle Global	16.00 %	2
Stall Sass Tex Mem Throttle Shared	16.00 %	2
Stall Sass Tex Mem Throttle Local	16.00 %	2
Stall Sass Tex Mem Throttle Other	16.00 %	2
Stall Sass Sleep	16.00 %	2
Stall Sass Sleep Sleep	16.00 %	2
Stall Sass Sleep Wait	16.00 %	2
Stall Sass Wait	16.00 %	2
Stall Sass Wait Sleep	16.00 %	2
Stall Sass Wait Wait	16.00 %	2
Stall Sass Other	16.00 %	2
Stall Sass Other Sleep	16.00 %	2
Stall Sass Other Wait	16.00 %	2

Pc Sampling Indicators

Roofline Analysis The chart shows the roofline model of the kernel. It shows the achieved performance (in FLOPs) and arithmetic intensity (FLOPs/Byte) of the kernel(s) relative to the peak FLOPs of the device(s) FP16 peak performance and FP16 of the FP64 peak performance. See the Roofline section for more details on roofline analysis.



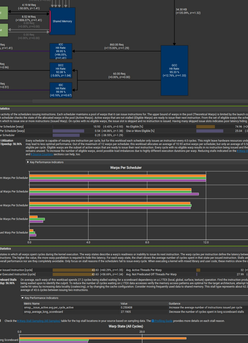
Memory Workload Analysis

Memory Chart

Memory Workload Analysis. The Memory Chart provides a detailed view of the memory units. Memory is shown including the following hardware units: Global Mem, unified by the unified L1/TEX cache, Local Mem, unified by the unified L1/TEX cache, Texture Mem, unified by the unified L1/TEX cache, Surfaces, unified by the unified L1/TEX cache, Shared Mem, Device Mem, Host Mem. Memory is broken down into its hierarchy of caches and memories. Each unit reports a wealth of throughput and utilization metrics. Hovering over the diagram's units and buses will provide more information on each metric.

Memory Throughput Chart in %	93.94 %	2,010.41 GB/s
Memory Throughput Chart in GB/s	93.94 %	2,010.41 GB/s
L1/TEX Hit Rate	0.00 %	0.00 GB/s
L2 Compression Success Rate	0.00 %	0.00 GB/s
L2 Compression Ratio	0.00	0.00 GB/s
L2 Compression Savings	0.00 %	0.00 GB/s
Kernel Level Cache Hit Rate	The memory access patterns for relevant kernels might not be optimal and caused an average of 2.77 cache lines (128B) across all 16384 relevant wavefronts. The results in SECTION Memory Workload Analysis and Memory Chart may be misleading. Consider fixing the memory access patterns first.	2.77

Key Performance Indicators


Memory Chart
Values
Transfer Size
Hashing
Unified Cache
Global Mem
Local Mem
Texture
Surfaces
Shared Mem
Unified Shared Memory
Tmem
Shared
Device
Host
L1/TEX Hit Rate
0 % HIT
0.00 GB/s
0.00 GB/s
L1 TEX
0 % HIT
0.00 GB/s
0.00 GB/s
Non-coherent Global
0 % HIT
0.00 GB/s
0.00 GB/s
Global
0 % HIT
0.00 GB/s
0.00 GB/s
Texture
0 % HIT
0.00 GB/s
0.00 GB/s
Surfaces
0 % HIT
0.00 GB/s
0.00 GB/s
Shared
0 % HIT
0.00 GB/s
0.00 GB/s
Local
0 % HIT
0.00 GB/s
0.00 GB/s
L2 Cache
0 % HIT
0.00 GB/s
0.00 GB/s
Device Memory
0 % HIT
0.00 GB/s
0.00 GB/s
L2 Fabric
0 % HIT
0.00 GB/s
0.00 GB/s
Sysmem Fabric
0 % HIT
0.00 GB/s
0.00 GB/s
Sysmem
0 % HIT
0.00 GB/s
0.00 GB/s
Device
0 % HIT
0.00 GB/s
0.00 GB/s
Host
0 % HIT
0.00 GB/s
0.00 GB/s
Peer
0 % HIT
0.00 GB/s
0.00 GB/s
Unified Shared Memory
0 % HIT
0.00 GB/s
0.00 GB/s
Tmem
0 % HIT
0.00 GB/s
0.00 GB/s
L2 Hit Rate
0 % HIT
0.00 GB/s
0.00 GB/s
L2 Compression
0 % HIT
0.00 GB/s
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L2 Compression Success Rate
0 % HIT
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L2 Compression Ratio
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L2 Compression Savings
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L2 Fabric
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Device Memory
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Sysmem Fabric
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Sysmem
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Peer
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Unified Shared Memory
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Tmem
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Device Memory
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L2 Fabric
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Sysmem Fabric
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Sysmem
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Unified Shared Memory
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Unified Shared Memory
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Unified Shared Memory
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0.00 GB/s
Tmem